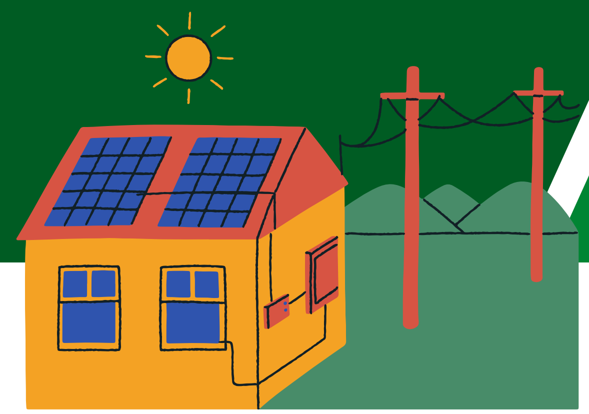
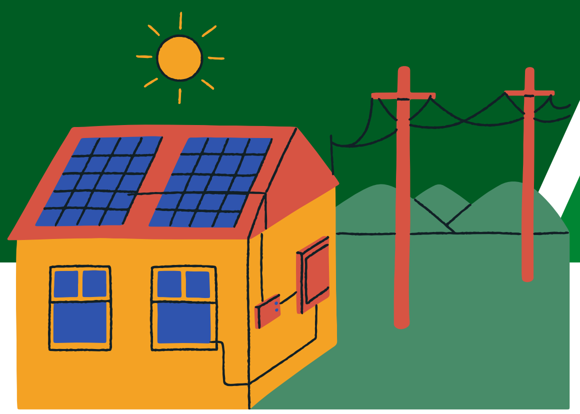


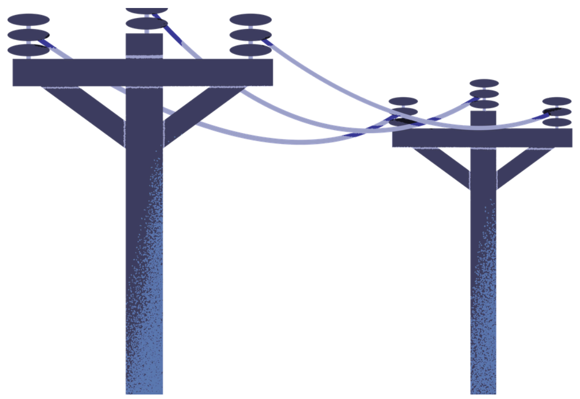
GROUP 1

# STRUCTURAL BREAKS, TREND DYNAMICS, AND FORECASTING PERFORMANCE OF PHILIPPINE ELECTRICITY DEMAND: EVIDENCE FROM 2019 - 2024

CLANOR | LAZONA



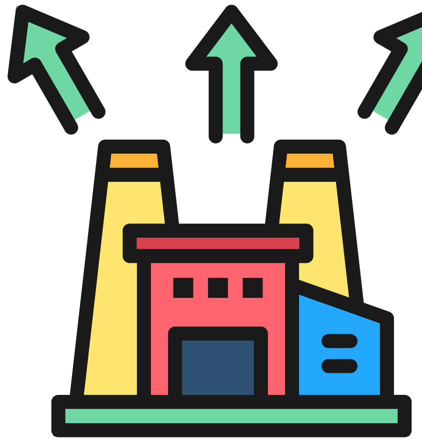
## INTRODUCTION



### ELECTRICITY DEMAND IN THE PHILIPPINES

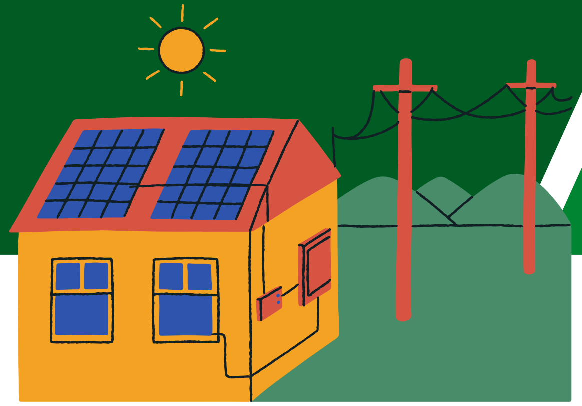
Key indicator of economic activity and development driven by:

**Population Growth**



**Industrial Expansion**

**Energy Policies**



# INTRODUCTION

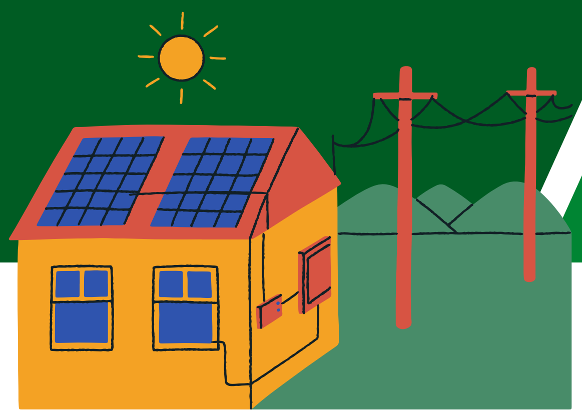
**Demand is not stable or linear.**
**Influenced by:**

Seasonality

Economic Disruptions

Structural Changes

- Events like COVID-19 caused sudden shifts (structural breaks
- Traditional models may produce inaccurate forecasts if breaks are ignored



# STATEMENT OF THE PROBLEM

1

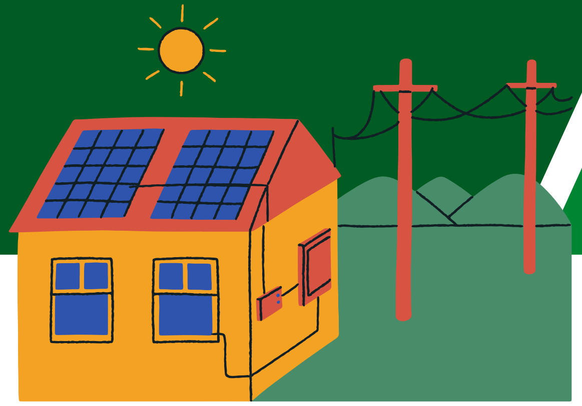
Does Philippine monthly electricity demand exhibit a statistically significant long-term trend after controlling for seasonality, and how does this trend differ across identified regimes?

2

Did significant structural breaks occur in Philippine electricity demand between 2019 and 2024, indicating regime shifts, and how do these breaks affect forecasting accuracy?

3

Is the deseasonalized and detrended electricity demand stationary, and how does accounting for volatility in the remainder component improve short-term and medium-term forecasting?



# OBJECTIVES OF THE STUDY

1. Decompose electricity demand into trend, seasonal, and remainder components using STL decomposition.

2. Test for statistically significant long-term trends using the Mann-Kendall test and Sen's slope estimator.

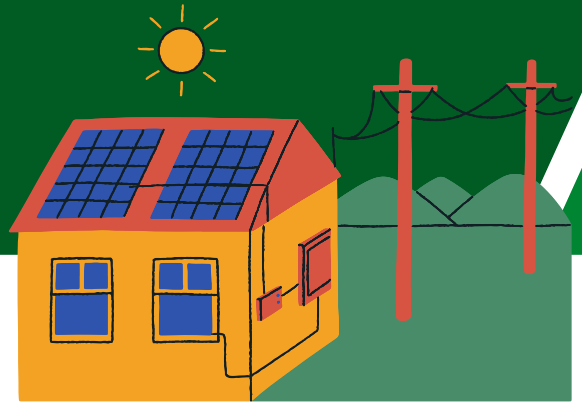
3. Identify multiple structural breakpoints using the Bai-Perron method.

4. Compare regime-specific statistics including mean, variance, volatility, and trend slope.

5. Test for stationarity using the Augmented Dickey-Fuller (ADF) test.

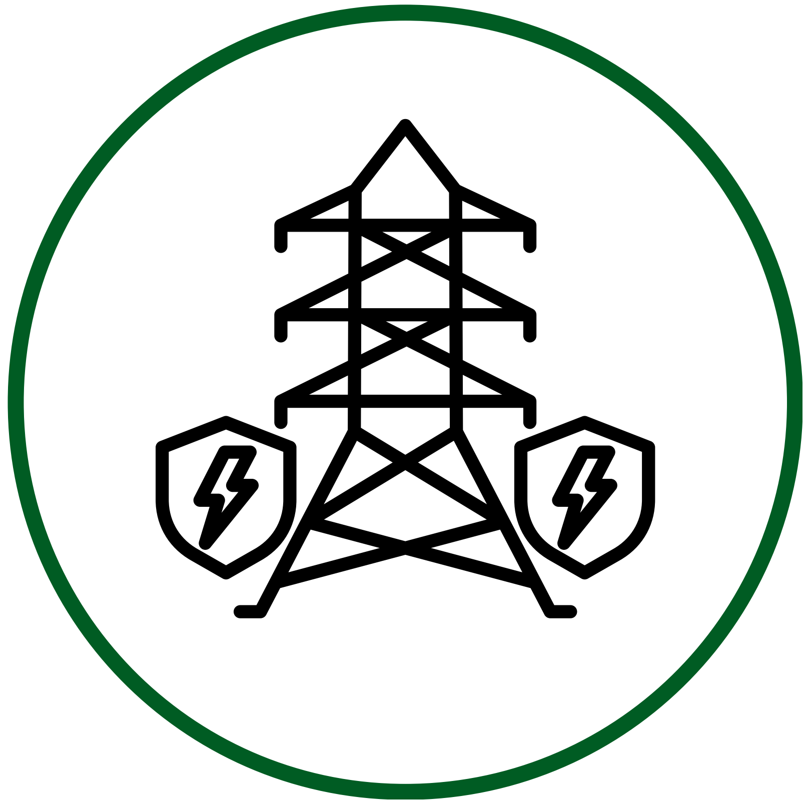
6. Evaluate forecasting performance across:
   - Standard SARIMA models
   - Break-adjusted ARIMA models
   - STL-based component forecasting
   - Prophet modeling

7. Compare forecasting accuracy using RMSE, MAE, and MAPE.



# SIGNIFICANCE OF THE STUDY

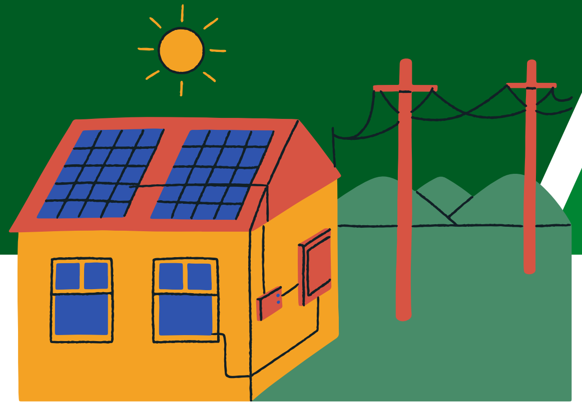
CAPACITY PLANNING



GRID STABILITY MANAGEMENT

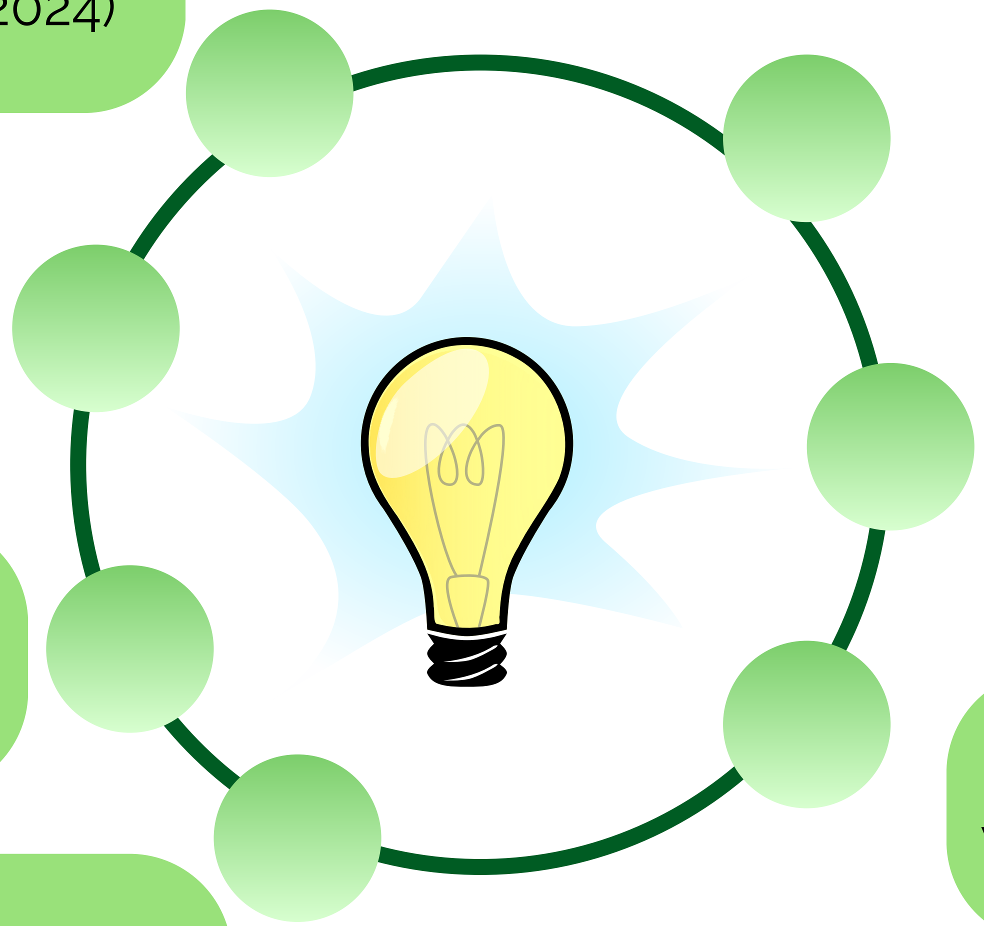
DEMAND FORECASTING FOR PROCUREMENT DECISIONS

LONG-TERM INFRASTRUCTURE INVESTMENT PLANNING

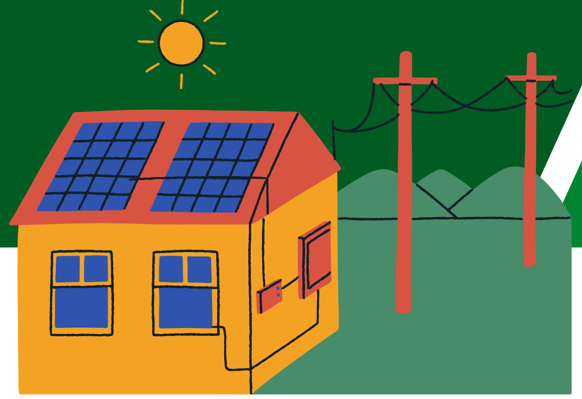


# SCOPE AND LIMITATION

- Monthly total electricity consumption in the Philippines(January 2019 – December 2024)
- Univariate time series analysis only
- Structural breaks detected endogenously using statistical tests
- Forecast evaluation based on out-of-sample validation within the study period



- Does not include exogenous macroeconomic variables(e.g., GDP, temperature, fuel prices)
- No long-term projections beyond the observed time horizon
- Results are confined to historical dynamics within the sample period



# METHODOLOGY

**Data & Diagnostics Raw Data**

**Time-Series STL Decomposition Trend Tests (MK + Sen)**

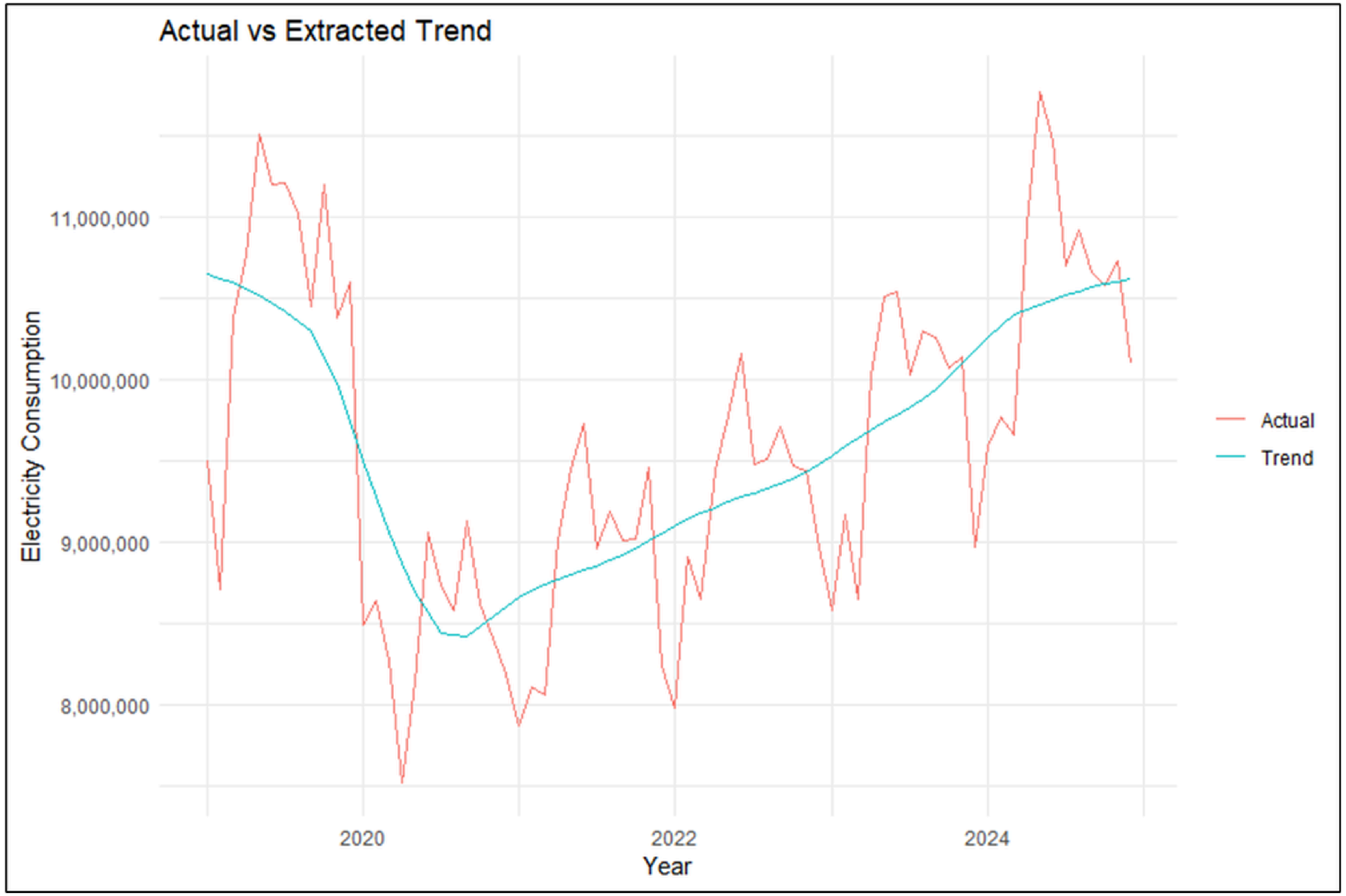
**Structural Analysis Bai-Perron Break Detection Regime Statistics ADF Stationarity**

**Forecasting & Evaluation**

SARIMA, ARIMA, STL, Prophet Model Comparison (RMSE, MAE, MAPE)

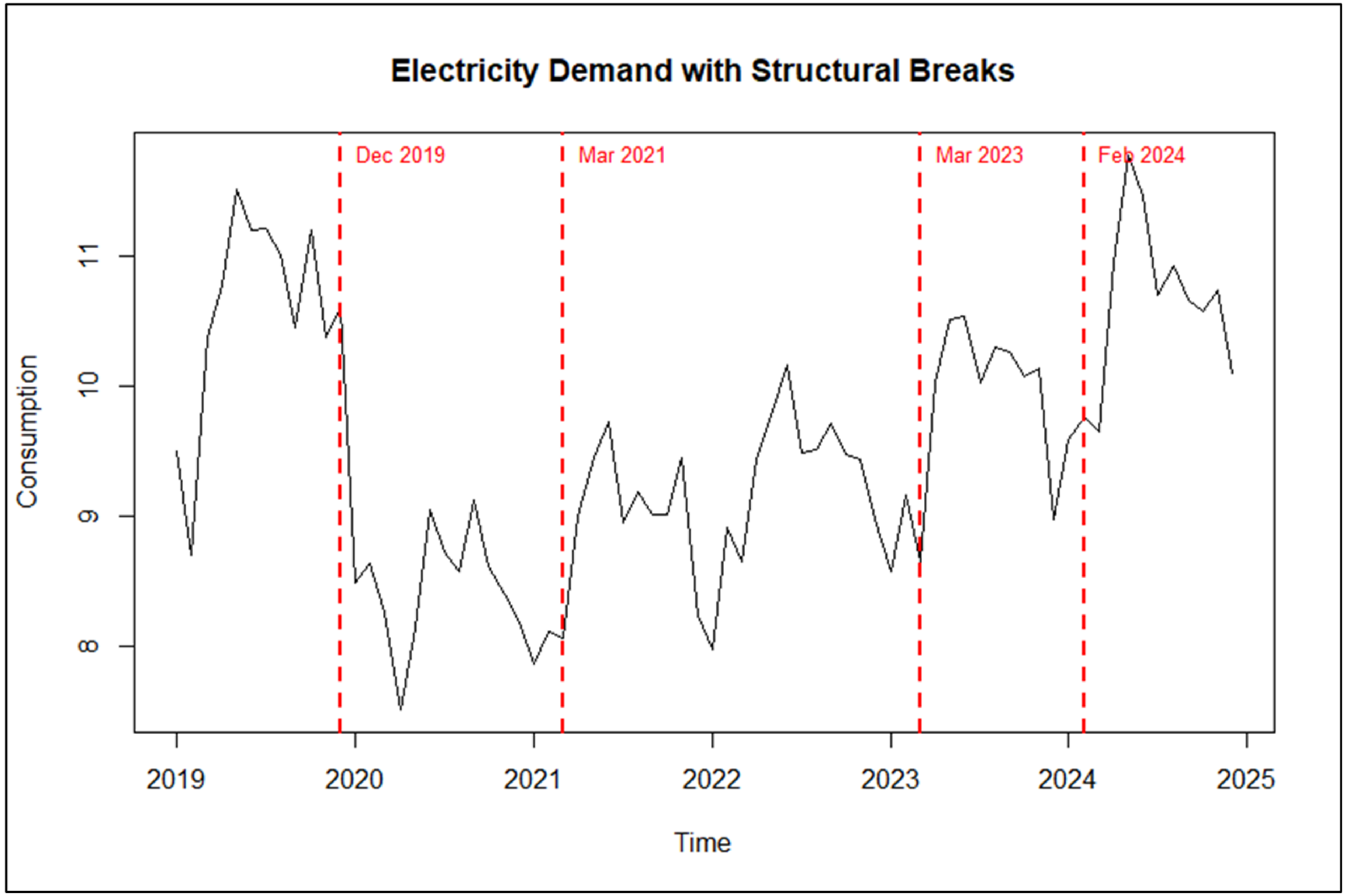
# RESULTS AND DISCUSSIONS

## TREND ANALYSIS (STL)

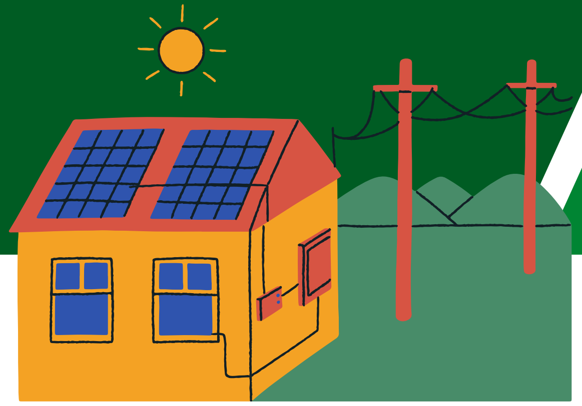



- Upward trend observed
- Mann–Kendall: significant ($p < 0.01$)
- Sen's slope: ~35,180/month
- ~422,000 annual increase

# STRUCTURAL BREAK ANALYSIS




- 4 breakpoints detected
- 5 distinct regimes



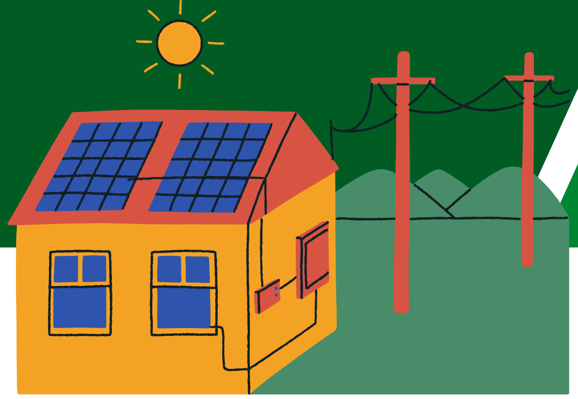
# RESULTS AND DISCUSSIONS

## REGIME-LEVEL MEAN COMPARISON

| Regime | Mean Consumption |
|---|---|
| Segment 1 | 10,579,638 |
| Segment 2 | 8,386,691 |
| Segment 3 | 9,166,021 |
| Segment 4 | 10,020,461 |
| Segment 5 | 10,757,970 |

# REGIME-LEVEL VARIANCE

| Regime | Variance |
| --- | --- |
| Segment 1 | 642,000,000,000 |
| Segment 2 | 186,000,000,000 |
| Segment 3 | 262,000,000,000 |
| Segment 4 | 201,000,000,000 |
| Segment 5 | 366,000,000,000 |



# RESULTS AND DISCUSSIONS

## REGIME-LEVEL REMAINDER VOLATILITY

| Regime | SD (Remainder) |
|---|---|
| Segment 1 | 607,918 |
| Segment 2 | 480,793 |
| Segment 3 | 214,156 |
| Segment 4 | 295,312 |
| Segment 5 | 306,094 |

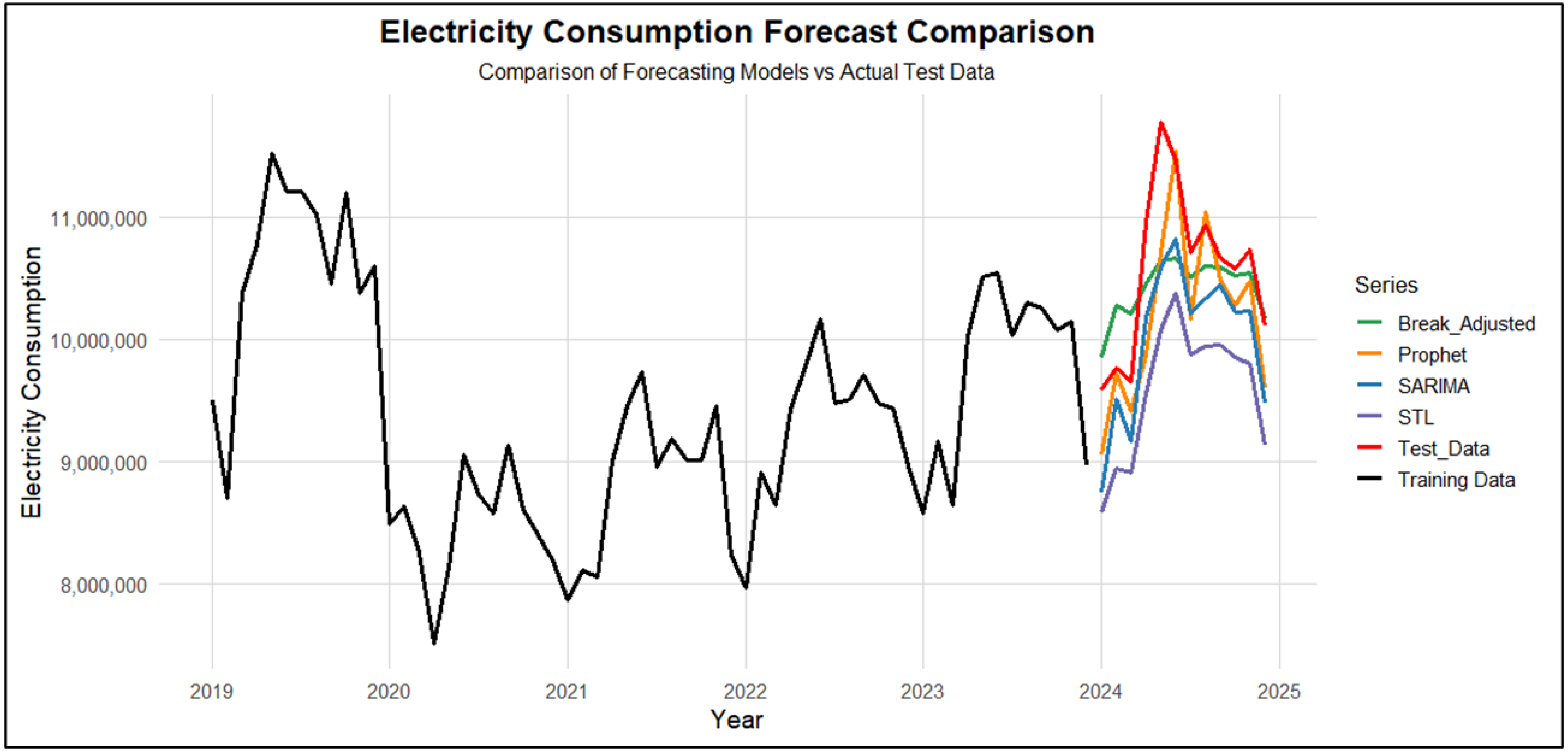
# RESULTS AND DISCUSSIONS

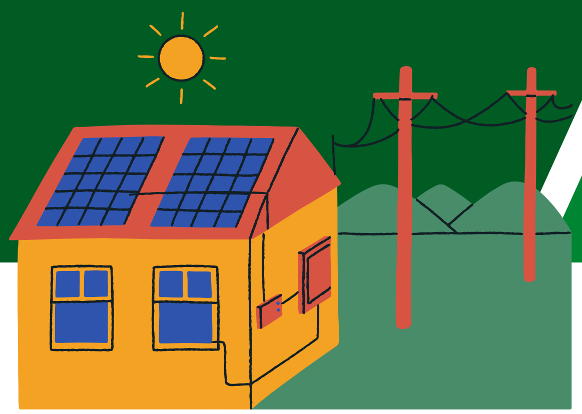
## STATIONARITY AND SHOCK PERSISTENCE

| Series | DF value | P-value | Stationarity |
|---|---|---|---|
| Original Series | −3.055 | 0.146 | Non-stationary |
| Trend Component | −3.271 | 0.083 | Non-stationary |
| Remainder Component | −4.520 | 0.01 | Stationary |

# RESULTS AND DISCUSSIONS

## FORECASTING MODELS

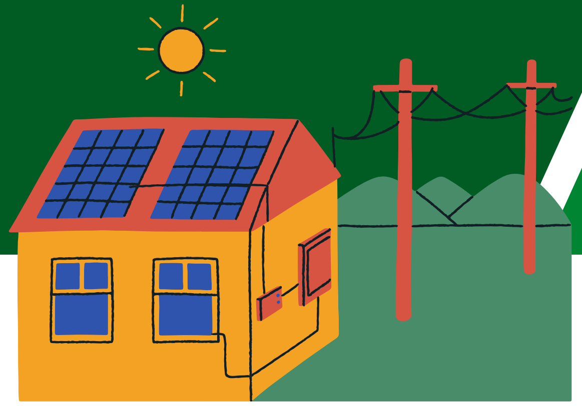




# RESULTS AND DISCUSSIONS

## FORECASTING PERFORMANCE

| Model | RMSE | MAE | MAPE |
|---|---|---|---|
| SARIMA | 637334.4 | 582950.7 | 5.472554 |
| Break-Adjusted ARIMA | 500174 | 388447.9 | 3.603301 |
| STL-Based | 1031012.1 | 991592.3 | 9.310244 |
| Prophet | 530191.7 | 409177.9 | 3.820314 |



# SUMMARY OF FINDINGS

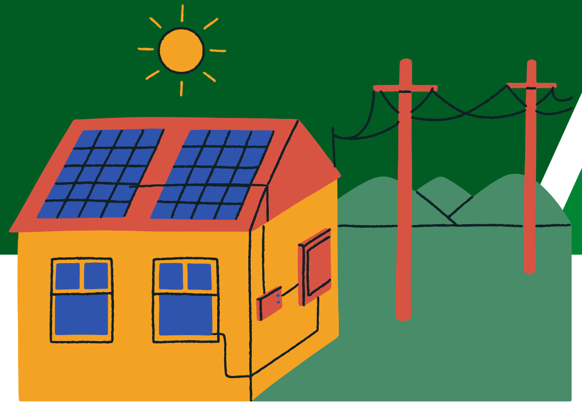
Philippine electricity demand shows a **significant long-term upward trend**

Demand evolves across **multiple structural regimes**, not a single pattern

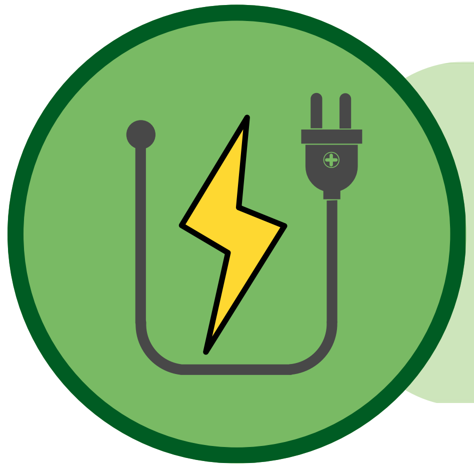
**Short-term shocks are temporary**, but long-term growth persists

Models that account for **structural breaks perform better**

**Break-adjusted ARIMA (1,0,0)(1,0,0)[12]** achieved the highest accuracy (lowest RMSE, MAE, MAPE)

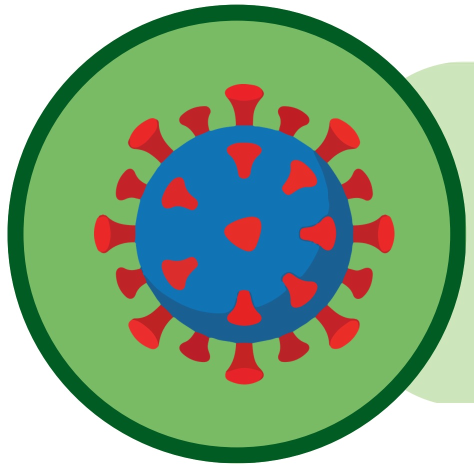


# CONCLUSION

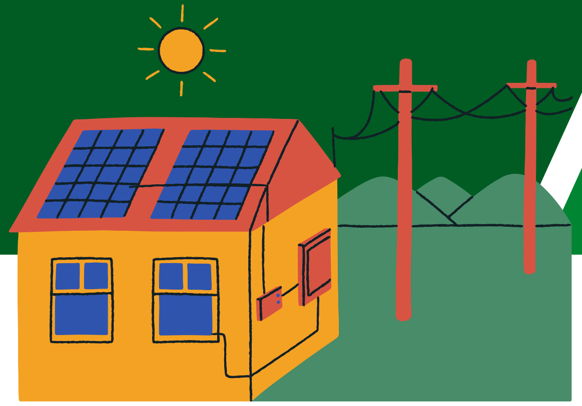


Philippine electricity demand from 2019-2024 is not stationary and structurally evolving, with changing trends and volatility over time

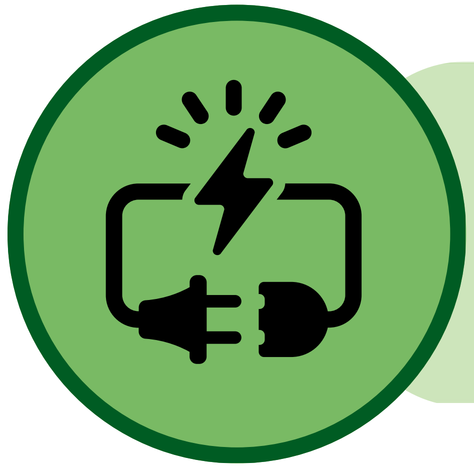
A significant upward trend is observed, with an average increase of about 35, 179 MWh per month



Four structural breaks divide the series into distinct regimes, reflecting shifts due to events such as the the COVID-19 pandemic

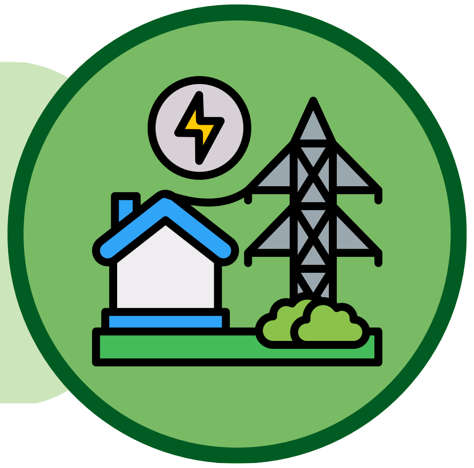


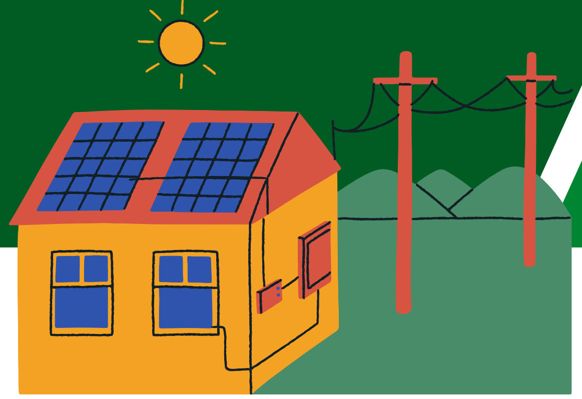
# CONCLUSION



Long-term movements are persistent, while short-term fluctuations are temporary and mean-reverting

The break-adjusted ARIMA model performs best, showing that accounting for structural breaks improves forecasting accuracy





# RECOMMENDATION

**Incorporate structural breaks** in forecasting models to improve accuracy.

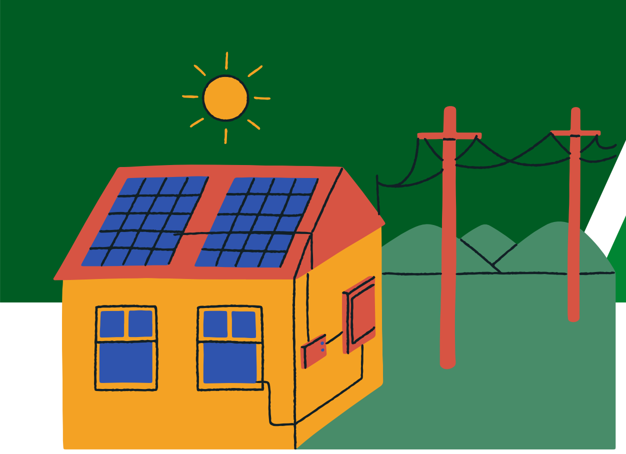
**Use hybrid modeling approaches** to capture both long-term and short-term patterns.

**Include external factors** (e.g., economic conditions, weather, policies) in future analyses.

**Extend the dataset** to better understand long-term demand behavior.

**Adopt adaptive forecasting systems** that update with new data.

**Explore advanced machine learning models** for improved performance.



GROUP 1

# STRUCTURAL BREAKS, TREND DYNAMICS, AND FORECASTING PERFORMANCE OF PHILIPPINE ELECTRICITY DEMAND: EVIDENCE FROM 2019 - 2024

CLANOR | LAZONA